Matthew Ngoy

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PROFESSIONAL SUMMARY

Computer science senior focused on cybersecurity, AI security research, and secure infrastructure; seeking a Summer 2027 internship.

EDUCATION

Kean University — Union, NJ

Bachelor of Science in Computer Science (Cybersecurity Option); GPA: 3.52 — *Expected May 2027*

- **Relevant Courses:** Ethical Hacking; Digital Forensics; Computer Security; Cryptography & Cryptanalysis; Data Structures; Analysis of Algorithms; Operating Systems; Database Management Systems.
- **Honors & Programs:** NSF IUSE Scholar (2x); CRA UR2PhD; CAHSI Travel Scholar; Dean's List; Upsilon Pi Epsilon; WiCyS Secretary; Young Envoys Scholarship.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, PHP, PowerShell
Security: OWASP Top 10, MITRE ATLAS, OSINT, Nmap
AI & LLM: Ollama, OpenAI API (fine-tuning), LLM-as-judge evaluation, Claude Code, Codex
Tools & Frameworks: Linux (Ubuntu, Kali), Docker, Git, MySQL, SQLite, Node.js, Arduino/ESP8266
Coursework Exposure: TypeScript, Wireshark, Metasploit, Hashcat, Autopsy

EXPERIENCE

Cybersecurity Research Intern — May 2026 – Present

Kean University Center for Cybersecurity — *Union, NJ / Wenzhou, China / Beijing, China*

- Investigated nonprofit cyber incidents through OSINT for the Non-Profit Cyber Incident Repository, classifying reports by CIA-triad impact, CISA sector, and TAG Cyber taxonomy.
- Designing solar-power and rack-energy monitoring for Mobile ACDC, the Center's portable teaching data center.
- Built an Arduino/ESP8266 smart-home lab and configured classroom computers for the Port Authority of NY/NJ STEaM Camp, preparing 30 laptops.
- Taught cryptography, programming, and IP addressing across 12 lessons at STEaM Camp, reaching 60 high school students.

Peer Mentor / Code Samurai — Jan – May 2025; Sep 2026 – Present

Kean University Department of Computer Science (Code Samurai) — *Union, NJ*

- Volunteered in Spring 2025 and returned as a paid mentor in Fall 2026, helping students work through Python, SQL, data structures, and debugging problems.

International Youth Delegate — Sep 2025

International Forum We, the Youth (IFWY) — *New York, NY / Seoul, Korea*

- Spearheaded project design and scope for SelectSolar, a proposal to divert solar panels from landfills and deliver low-cost, energy-efficient power solutions to educational and non-profit institutions.
- Selected for a fully funded fellowship at Cornell Tech, then voted 1 of 25 North American delegates for the IFWY Final Conference in Seoul, collaborating on international environmental policy and sustainable solutions.

PROJECTS

KEAN CTF 2027 | *Python, Docker, CTFd, Cloudflare Tunnel, GitHub Actions* — Sep 2026 – Present

- Co-developing a self-hosted college capture-the-flag competition with AI assistance and two teammates, using Docker-isolated challenges.
- Co-developed challenges and solve scripts with AI assistance, including defenses against blind AI solvers.
- Used AI-assisted development to automate challenge validation in CI and correct an unintended flag leak and broken hints during a player audit.

LLM Red-Team Harness | *Python, SQLite, Ollama* — June 2026

- Developed with AI assistance a pipeline for prompt-injection, jailbreak, and data-leak tests, recording model-judge verdicts and transcripts in SQLite and JSONL.

RESEARCH & PRESENTATIONS

Independent Study, Red-Teaming HadAgent | *Advisor: Dr. Boyang Li* — Sep 2026 – Present

- Directed questions about verification guarantees and interpreted AI-designed and AI-run tests of a copied HadAgent prototype, identifying score-validation and model-execution gaps in local testing.

Debate Coach LLM Evaluation | *CRA UR2PhD; JavaScript, Node.js, OpenAI API* — Feb – Apr 2026

- Helped fine-tune and evaluate LLM debate coaches with two teammates, sharing citation checks, participant interviews, and presentation; used an AI-assisted audit to investigate source reliability.